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Exp5: Develop a web application to fetch real-time weather data for a city. Use AJAX & a public weather API to retrieve information. Display temperature, humidity, and weather conditions dynamically.

index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Weather App - Real-time Data</title>
    <!-- Bootstrap CSS -->
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
    <link rel="stylesheet" href="style.css">
</head>
<body class="bg-light">

    <div class="container py-5">
        <div class="row justify-content-center">
            <div class="col-md-6">
                <div class="card shadow-sm">
                    <div class="card-header bg-primary text-white text-center">
                        <h3 class="mb-0">Weather Forecast</h3>
                    </div>
                    <div class="card-body">
                        <div class="input-group mb-3">
                            <input type="text" id="cityInput" class="form-control" placeholder="Enter city
name..." aria-label="City name">
                            <button class="btn btn-primary" type="button"
id="getWeatherBtn">Search</button>
                        </div>

                        <div id="loader" class="text-center d-none">
                            <div class="spinner-border text-primary" role="status">
                                <span class="visually-hidden">Loading...</span>
                            </div>
                        </div>

                        <div id="weatherResult" class="d-none">
                            <hr>
                            <div class="text-center mb-4">
                                <h2 id="displayCity" class="fw-bold text-uppercase">City Name</h2>
                                <h1 id="displayTemp" class="display-1 fw-bold">25°C</h1>
                                <p id="displayCondition" class="lead text-muted">Cloudy</p>
                            </div>
```

```
                <div class="row text-center mt-4">
                    <div class="col-6">
                        <div class="p-3 border rounded bg-white">
                            <h6 class="text-muted">Humidity</h6>
                            <h4 id="displayHumidity">60%</h4>
                        </div>
                    </div>
                    <div class="col-6">
                        <div class="p-3 border rounded bg-white">
                            <h6 class="text-muted">Wind Speed</h6>
                            <h4 id="displayWind">12 km/h</h4>
                        </div>
                    </div>
                </div>
            </div>

            <div id="errorMessage" class="alert alert-danger d-none mt-3" role="alert">
                City not found. Please try again.
            </div>
        </div>
    </div>
    <p class="text-center text-muted mt-4 small">Data provided by Open-Meteo API</p>
  </div>
 </div>
</div>

<!-- jQuery and Bootstrap JS -->
<script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
<script src="script.js"></script>
</body>
</html>
```

style.css:

```
body {
    background: linear-gradient(135deg, #f5f7fa 0%, #c3cfe2 100%);
    min-height: 100vh;
}

.card {
    border: none;
    border-radius: 15px;
```

```
}

.card-header {
    border-top-left-radius: 15px !important;
    border-top-right-radius: 15px !important;
}

#displayTemp {
    color: #0d6efd;
}

#weatherResult {
    transition: all 0.3s ease-in-out;
}
```

script.js:

```
$(document).ready(function() {
    $("#getWeatherBtn").click(function() {
        const city = $("#cityInput").val().trim();
        if (city === "") {
            showError("Please enter a city name.");
            return;
        }

        fetchWeather(city);
    });

    // Enter key support
    $("#cityInput").keypress(function(e) {
        if (e.which == 13) {
            $("#getWeatherBtn").click();
        }
    });

    function fetchWeather(city) {
        // Reset UI
        $("#weatherResult").addClass("d-none");
        $("#errorMessage").addClass("d-none");
        $("#loader").removeClass("d-none");

        // Step 1: Get location key using city search
        $.ajax({
            url: `/api/search?q=${encodeURIComponent(city)}`,
```

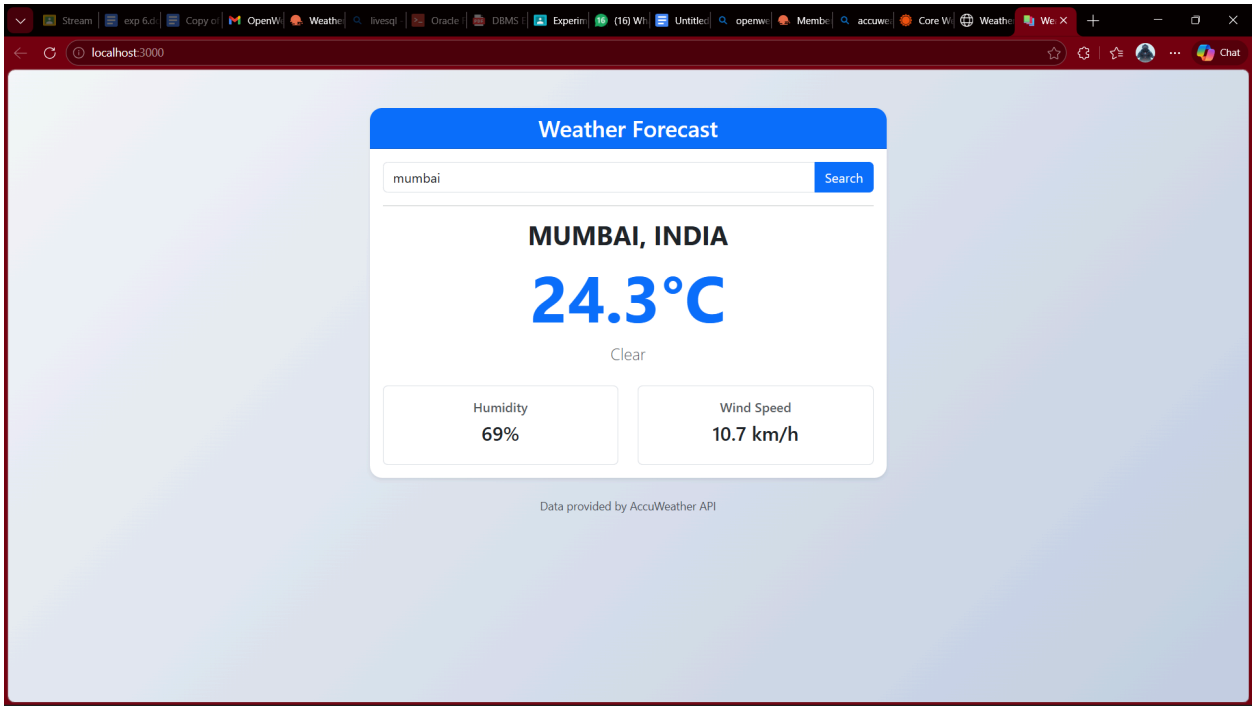
```
        method: "GET",
        success: function(locationData) {
            if (locationData && locationData.length > 0) {
                const location = locationData[0];
                const locationKey = location.Key;
                const cityName = `${location.LocalizedName}, ${location.Country.LocalizedName}`;

                // Step 2: Fetch weather data using location key
                $.ajax({
                    url: `/api/weather/${locationKey}`,
                    method: "GET",
                    success: function(weatherData) {
                        displayWeather(cityName, weatherData[0]);
                    },
                    error: function() {
                        showError("Failed to fetch weather data.");
                    },
                    complete: function() {
                        $("#loader").addClass("d-none");
                    }
                });
            } else {
                showError("City not found. Please check the spelling.");
                $("#loader").addClass("d-none");
            }
        },
        error: function(xhr) {
            showError("Failed to fetch city data. Server connection issue.");
            $("#loader").addClass("d-none");
        }
    });
}

function displayWeather(cityName, data) {
    $("#displayCity").text(cityName);
    $("#displayTemp").text(`${data.Temperature.Metric.Value}°C`);
    $("#displayCondition").text(data.WeatherText);
    $("#displayHumidity").text(`${data.RelativeHumidity}%`);
    $("#displayWind").text(`${data.Wind.Speed.Metric.Value} ${data.Wind.Speed.Metric.Unit}`);

    $("#weatherResult").removeClass("d-none");
}

function showError(msg) {
```

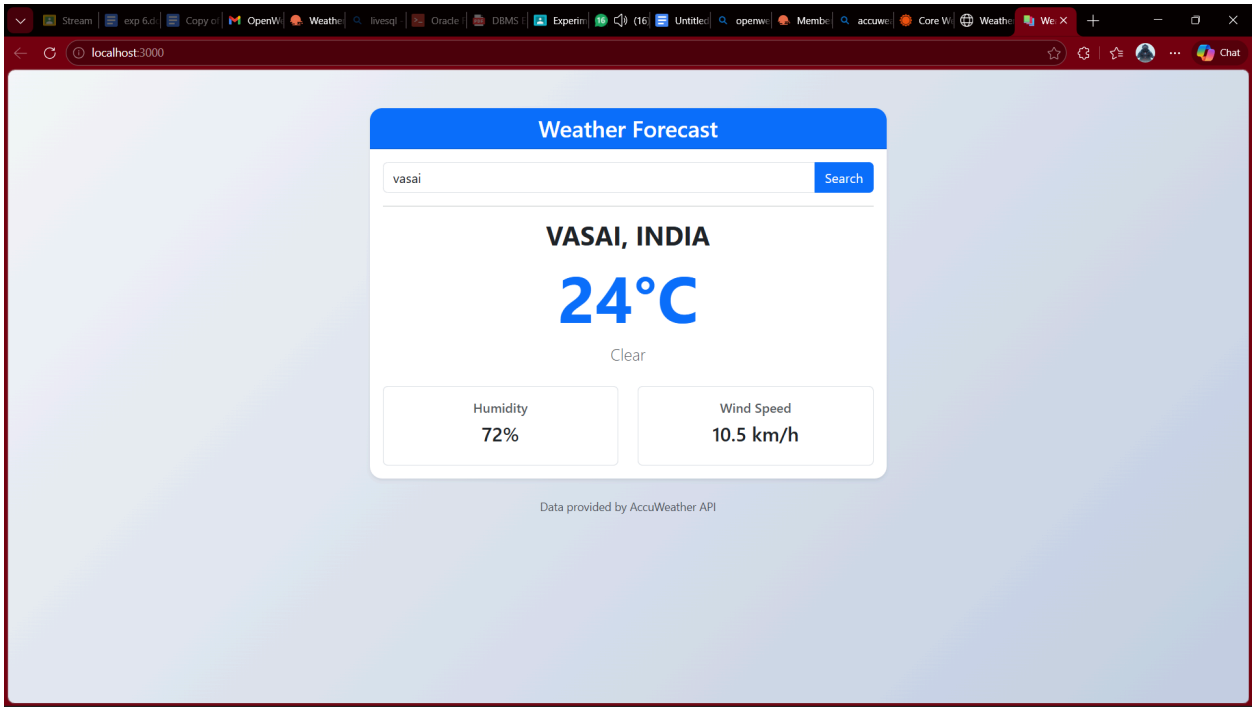
```
        $("#errorMessage").text(msg).removeClass("d-none");
        $("#loader").addClass("d-none");
        $("#weatherResult").addClass("d-none");
    }
});
```

server.js:

```
const express = require('express');
const axios = require('axios');
const cors = require('cors');
const path = require('path');

const app = express();
const PORT = 3000;

// Middleware
app.use(cors());
app.use(express.json());
app.use(express.static(path.join(__dirname)));

// AccuWeather API key
const ACCUWEATHER_API_KEY = "zpka_3cb7f27f24354bec960a50f3c9c0b820_404a373c";
const BASE_URL = "https://dataservice.accuweather.com";

// Route to search for a city
app.get('/api/search', async (req, res) => {
    try {
        const { q } = req.query;
        if (!q) {
            return res.status(400).json({ error: "City name is required" });
        }

        const response = await axios.get(`${BASE_URL}/locations/v1/search`, {
            params: { q },
            headers: { Authorization: `Bearer ${ACCUWEATHER_API_KEY}` }
        });

        res.json(response.data);
    } catch (error) {
        console.error("Search error:", error.message);
        res.status(500).json({ error: "Failed to search city" });
    }
```

```
});

// Route to get current weather
app.get('/api/weather/:locationKey', async (req, res) => {
    try {
        const { locationKey } = req.params;

        const response = await axios.get(`${BASE_URL}/currentconditions/v1/${locationKey}`, {
            params: { details: true },
            headers: { Authorization: `Bearer ${ACCUWEATHER_API_KEY}` }
        });

        res.json(response.data);
    } catch (error) {
        console.error("Weather error:", error.message);
        res.status(500).json({ error: "Failed to fetch weather data" });
    }
});

// Serve the HTML file
app.get('/', (req, res) => {
    res.sendFile(path.join(__dirname, 'index.html'));
});

app.listen(PORT, () => {
    console.log(`Server running at http://localhost:${PORT}`);
    console.log("Open http://localhost:3000 in your browser");
});
```


localhost:3000
Weather Forecast
mumbai
Search
MUMBAI, INDIA
24.3°C
Clear
Humidity
69%
Wind Speed
10.7 km/h
Data provided by AccuWeather API


localhost:3000
Weather Forecast
vasai
Search
VASAI, INDIA
24°C
Clear
Humidity
72%
Wind Speed
10.5 km/h
Data provided by AccuWeather API